SIMATS ENGINEERING

Lab Programs

COURSE CODE: ECA1009 **COURSE NAME: Microprocessors & Microcontrollers**

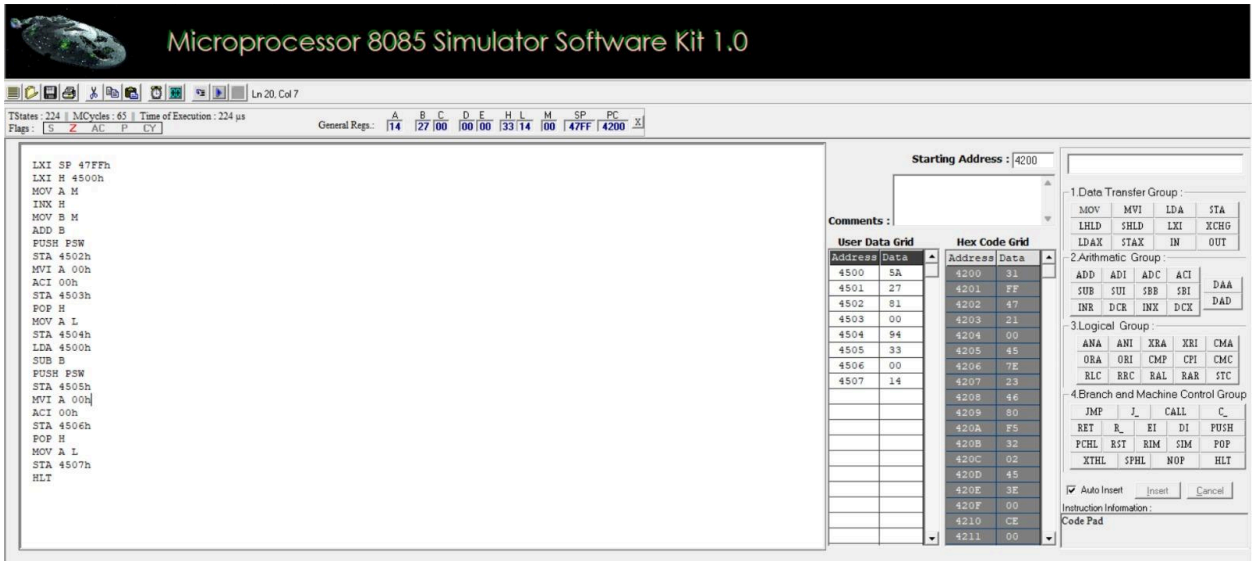
Program A – Develop and Verification of 8-Bit Addition and Subtraction Using Intel 8-Bit Microprocessor Assembly Language Programming

Aim: To perform arithmetic operations on 8-bit numbers using the 8085 microprocessor and store the results in memory.

Algorithm:

1. Initialize the stack pointer and load the HL register pair with the first data address.
2. Move the first number into the accumulator and increment the memory pointer.
3. Load the second number into register B.
4. Perform addition and store the result at memory location 4502H.
5. Perform subtraction and store the result at memory location 4504H.
6. Store the carry/borrow status at the specified memory locations and halt the program.

Execution:



Result: The 8085 microprocessor program was executed successfully, and the arithmetic results were stored in the specified memory locations.